

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460946.896 +/- 0.028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.028
Transit depth (Rp/Rs)^2	3.12 +/- 1.0 [%]
Orbital Inclination (inc)	86.3 +/- 1.41
Airmass coefficient 1 (a1)	0.956 +/- 0.011
Airmass coefficient 2 (a2)	0.03 +/- 0.4
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	None
Optimal Aperture	3.4
Optimal Annulus	10.34
Transit Duration (day)	0.062 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.028 vs 0.14195069566578392 (deviation 0.98σ)
Duration fit vs published	0.062 d vs 0.0875 d (deviation 0.63σ)
Mid-time O–C	-12.8 min
Depth SNR	5.0
Residual scatter	1.15 %

Light curve

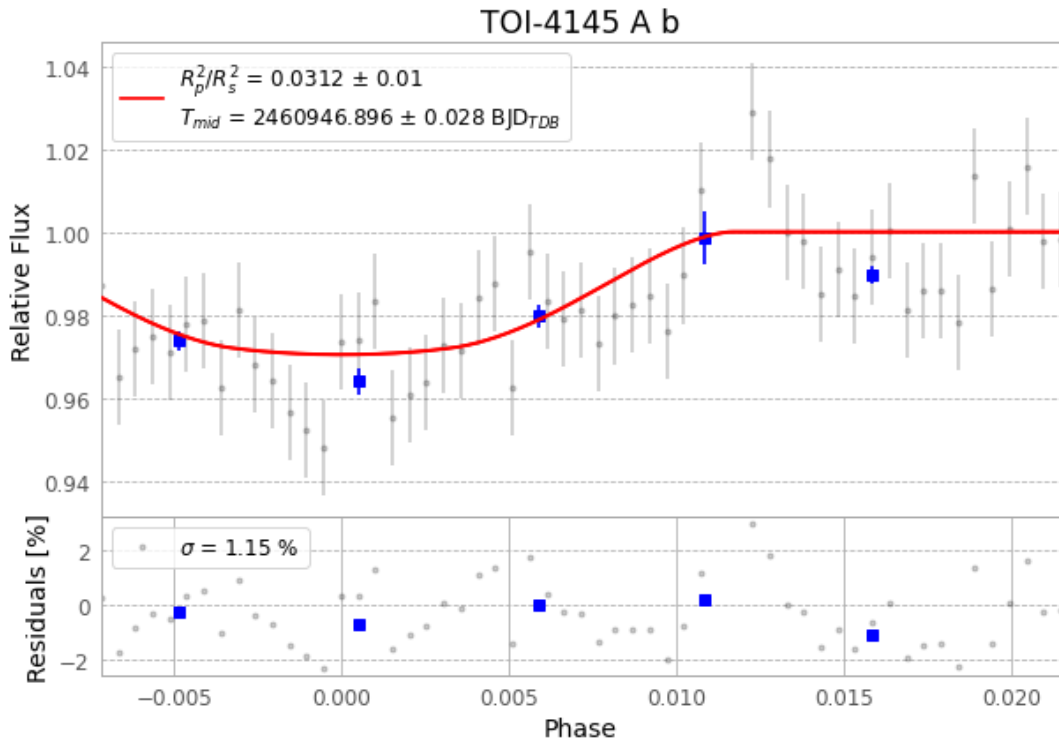


Figure 1 — FinalLightCurve_TOI-4145 A b_2025-09-28.png (SHA-256 b8d30b3065cdb1c7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [413, 286], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3374f813f100216a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

57 FITS frames (37 MB), TOI4145250928084516.FITS → TOI4145250928113316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4145-250928.log (0efc839cb81d...), FinalLightCurve_TOI-4145 A b_2025-09-28.png (b8d30b3065cd...), FinalLightCurve_TOI-4145 A b_2025-09-28.csv (b7d99792d432...)

Compute resources

Wall time 126.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 00:23Z.